# L — User Manual

ZX Spectrum 128K (LSpectrum), Commodore 64 (L64) and Atari 800XL / 130XE (LTari) on the QMTECH Cyclone IV board. Release of 2 November 2026.

## 1. Overview

L is one FPGA board that becomes one of three classic home computers. At power-up it shows the **machine menu**; you choose a machine, and the board becomes that computer, with its original ROMs and chips. Programs come from a microSD card: an on-screen file browser loads tapes, disks, cartridges and snapshots. Sound comes from the audio jack and, if you like, from Bluetooth headphones or a speaker.

| Machine | Computer | Files on the SD card |
|---|---|---|
| **1 LSpectrum** | ZX Spectrum 128K (also runs 48K programs) | .tap .tzx (tapes), .z80 (snapshots) |
| **2 L64** | Commodore 64 (PAL) with 1541 disk drive | .d64 (disks), .prg .t64 (programs), .tap (tapes), .a64 (snapshots) |
| **3 LTari 800** | Atari 800XL, 64 KB (PAL) | .atr (disks), .xex (programs), .car (cartridges), .a8s (snapshots) |
| **4 LTari 130** | Atari 130XE, 128 KB (PAL) | as LTari 800 |

> The board set comes without games, other programs and the Commodore 64 ROMs: get them yourself (Installation Guide, section 4 "Software not included").

## 2. Connections

| Connector | Connect | Notes |
|---|---|---|
| VGA (DB15) | Monitor | Starts at 640 x 480, 60 Hz (L64: 800 x 600, 60 Hz), which every monitor accepts. F8 switches to the machine's own 50 Hz if the monitor supports it. |
| DB9-1 (J4), DB9-2 (J6) | Joysticks | Atari / Commodore type (switches to ground): 1 up, 2 down, 3 left, 4 right, 6 fire, 8 ground. What each socket is in each machine: section 6. |
| 3.5 mm jack (J3) | Amplifier or active speakers | Line level, mono on both channels. Not for headphones directly. |
| Keyboard module (J5) | USB keyboard | A normal PC keyboard; the special keys are listed in this manual. |
| microSD socket (J7) | SD card | FAT32 or FAT16, not exFAT. Can be changed at any time. |

| Connector | Connect | Notes |
|---|---|---|
| Bluetooth | Headphones or speaker | Paired with F3 (section 5); the ESP32 remembers the device. |
| 5 V input | Power supply | Powers the core board, the I/O board and the ESP32. |

## 3. The machine menu

The menu appears at every power-up. It needs only the monitor and the keyboard or a joystick.

| Key | What it does |
|---|---|
| **Up / Down** (keypad 8 / 2) | Moves the highlight |
| **Enter** (keypad Enter) | Starts the highlighted machine |
| **1** ... **4** | Starts that machine at once |
| Joystick (either socket) | Up / down moves, fire starts |
| **F3** | Bluetooth audio menu (section 5), already in the menu, before any machine runs |
| **Ctrl + Alt + Del** in a machine | Back to this menu; the highlight is on the machine you left. That machine's state is lost. Bluetooth stays connected. |

If a machine cannot start (its part of the flash is missing or damaged), the menu comes back with red lines **Could not start**: install `LUPDATE.BIN` again (Installation Guide).

## 4. Keys in every machine

| Key | What it does |
|---|---|
| **F12**, keypad **/** or **NumLock** | Opens / closes the SD card file browser (section 4.1) |
| **F2** | Saves a snapshot of the whole machine into the browser's current folder (asks for a name) |
| **F3** | Bluetooth audio menu (section 5) |
| **F4** | Joystick menu (section 6) |
| **F8** | Swaps the video between 60 Hz and the machine's own 50 Hz. After power-up and after the menu it is always 60 Hz. If the monitor shows nothing, press F8 again. |
| **Print Screen** | Saves the picture as SCRNnnnn.BMP in the browser's current folder |
| **Ctrl + Alt + Del** | Back to the machine menu (acts when the keys are released). Starting the machine again from there is a complete restart. |

Short messages (for example **Speed: turbo**) appear for a few seconds on the bottom line of the picture. While a menu or the browser is open, the machine sees no keys.

### 4.1 The SD card file browser

The browser lists folders first, then the files the machine can use (up to 320 per folder), white on blue over the picture.

| Key (browser open) | What it does |
|---|---|
| **Up / Down**, keypad 8 / 2, F9 / F10 | Moves the selection (held: repeats) |
| **Left / Right**, keypad 4 / 6, Page Up / Page Down | One page up / down |
| **Enter**, keypad Enter, F11 | Opens the folder, or loads the file (what that means: in each machine's section) |
| **Esc**, **Backspace** | Up one folder; in the top folder closes the browser |
| **F12** | Closes the browser |
| **F2** | Saves a snapshot into this folder |

When a name is asked for (snapshot, or a tape recording), type up to 8 letters or digits; **Backspace** deletes, **Enter** saves, **Esc** cancels. A name that already exists is refused.

## 5. Bluetooth audio (F3)

The sound of every machine also goes through the ESP32 to Bluetooth headphones or a speaker. The F3 menu works in the machine menu and in every machine.

1. Put the headphones or the speaker into pairing mode.
2. Press **F3**. The second line shows the state (**Not connected** / **Connected: name**).
3. Choose **Scan for devices** and press **Enter**. After a few seconds the devices found are listed.
4. Move to your device and press **Enter**. The line changes to **Connected: name**.
5. Press **F3** or **Esc** to close the menu.

| Menu line | What it does |
|---|---|
| Scan for devices | Searches for Bluetooth audio devices (drops the current connection) |
| Disconnect | Ends the connection |
| Forget the device | Ends it and does not reconnect by itself any more |
| PIN | Enter, type the code, Enter (empty = automatic: 0000, 1234, 8888, 1111 are tried in turn) |
| Volume | Left / Right |
| Sound | Enter swaps between the machine's sound and a test tone (to check the Bluetooth side alone) |
| Audio: short / skip / fix | Counters for service: should stay small; "skip" should stay 0 |

The ESP32 remembers the device and reconnects by itself when both are switched on. The connection stays when you change machines. Bluetooth adds about 0.2 s of delay: fine for music, noticeable in fast games. If the menu says **No answer from the ESP32**, check that the ESP32 board is plugged in.

## 6. Joysticks (F4)

F4 opens the joystick menu: Up / Down, **Enter** selects, F4 or Esc closes. DB9-1 is the socket J4, DB9-2 the socket J6.

| Machine | Choices (the first is the default) |
|---|---|
| LSpectrum | **Kempston** on both sockets (most games) — **Kempston + Cursor** (DB9-2 = keys 5 6 7 8 and 0 for fire: Cursor, Protek, AGF games) — **Interface 2** (DB9-1 = Sinclair 1, keys 6-0; DB9-2 = Sinclair 2, keys 1-5). Back to Kempston when the machine is started again. |
| L64 | **DB9-1 = control port 2, DB9-2 = port 1** (most one-player games use port 2) — swapped — both sockets on port 2. |
| LTari | **DB9-1 = stick 0** (player 1 in most games), DB9-2 = stick 1 — swapped. |

## 7. LSpectrum (ZX Spectrum 128K)

| Key | What it does |
|---|---|
| **F1** held while the Spectrum starts | Starts the second ROM (48K BASIC in the public release) instead of the 128K ROM. Hold F1 while you choose LSpectrum in the machine menu (or while switching on). |
| **F5** | Memory contention and floating bus on (default) / off. Timing-critical games and border effects need it on. |
| **F6** | Tape loading and saving in turbo (default, 8 times faster) / normal speed |
| **F7**, keypad * | Stops the tape completely (back to normal speed) |
| **F11**, keypad 5 | Pauses / continues the tape |
| Keypad **-** | Goes back one tape block (while paused) |
| **Page Up / Page Down** | Moves the frame interrupt 1/8 line (50 Hz mode, browser closed): lines up border effects with the picture |
| Esc / Backspace | BREAK / DELETE on the Spectrum (browser closed) |

### Loading a game

1. Make the Spectrum wait for a tape: **Tape Loader** in the 128K menu, or `LOAD ""` and Enter.
2. Press **F12**, choose the .tap or .tzx file and press **Enter**. The tape plays; in turbo the Spectrum runs 8 times faster until the tape ends.

3. Multi-load games: when the bottom line says **Tape stopped (5=go on)** and the game asks for the next part, press keypad **5** (or F11).
4. A loading error: keypad **5** pauses, keypad **-** goes back one block per press, start `LOAD ""` again and press keypad 5. If a game's own loader fails in turbo, press **F6** and load again at normal speed.

### Saving to tape

1. Press **F12**, go into a folder and choose the first line **[Save to this folder]**.
2. Type a name (or keep **SAVEnnnn**) and press **Enter**: recording starts.
3. Save on the Spectrum as often as you like ( `SAVE "name"` ...): every block goes into this one .TAP file.
4. Press **F12** to stop. Load it later like any tape.

### Snapshots (.z80)

**F2** stops the Spectrum and asks for a name; NAME.Z80 is saved and the Spectrum runs on. To continue later: F12, choose the .z80, Enter — no LOAD is needed. Snapshots from emulators (48K and 128K) work too.

### Border effects

Stripes in the border (for example in Aquaplane) line up only at 50 Hz: press **F8**, load the game, and if needed move them with **Page Up / Page Down**. The setting returns to the default when the machine is started again.

## 8. L64 (Commodore 64)

The first start installs the C64 ROMs from the SD card (see the Installation Guide).

| Key | What it does |
|---|---|
| **F5** | Sound chip SID 6581 (default) / 8580 |
| **F6** | Loading in turbo (default: up to 4 times faster while the tape or the disk motor runs) / normal |
| **F7**, keypad * | Stops the tape |
| **F11**, keypad 5 | Pauses / continues the tape |
| **Alt + F1 ... F8** | The C64's own F1 ... F8 keys (Alt + F2 / F4 / F6 / F8 = SHIFT + F1 / F3 / F5 / F7) |

### Loading

- **Disk (.d64)**: F12, choose the .d64, **Enter**. The disk goes into drive 8 and `LOAD"*",8,1` is typed for you. When **READY.** appears, type `RUN` and Return. For the directory: press Esc (RUN/STOP), then `LOAD"$",8` and `LIST`. When a game asks for another disk, choose it with F12.

- **Program (.prg, .t64)**: F12, choose it, **Enter**. A BASIC program starts by itself; for machine code the bottom line shows the address (for example $C000: `SYS 49152`).
- **Tape (.tap)**: type `LOAD` and Return (PRESS PLAY ON TAPE), then F12, choose the .tap, **Enter**.
- **Saving to tape**: F12, **[Save to this folder]**, a name, Enter; then `SAVE "name"` on the C64; F12 stops.
- **Snapshots (.a64)**: F2 saves (best while the disk drive is idle); F12, the .a64, Enter continues.

### The C64 keyboard

| PC key | C64 key |
|---|---|
| Letters, digits, Return, Space, , . / | The same keys |
| - = [ ] ; ' ` | + - @ * : ; ← (the keys at the same places on the C64) |
| \ | = |
| Insert / End / Home | £ / ↑ / CLR HOME |
| Backspace, Delete | INST DEL |
| Esc | RUN/STOP |
| Tab, Ctrl | CTRL |
| Windows key | C= (Commodore key) |
| Page Up (browser closed) | RESTORE |
| Cursor keys | Cursor keys (Left and Up add SHIFT, as on the C64) |
| Alt + F1 ... F8 | F1 ... F8 |

The key positions follow a US keyboard. Most games want the joystick in **control port 2**: that is DB9-1 by default. Cartridges are not supported.

## 9. LTari (Atari 800XL / 130XE)

| Key | What it does |
|---|---|
| **Alt + F2 / F3 / F4** | The Atari's START / SELECT / OPTION (held as long as the keys) |
| **Alt + F5** | The Atari's RESET key (warm start: BASIC keeps its program) |
| **Alt + F6** | HELP |
| **Pause**, **Alt + F7** | BREAK |
| **Shift + Enter** on an .atr | Changes the disk in drive D1: without restarting ("insert disk 2") |

### Loading

- **Disk (.atr)**: F12, choose it, **Enter**. Three lines ask **How do you want to load it: <F1> No BASIC <F2> with BASIC**. Most games, especially for the old 400 / 800, want **F1**; DOS disks

with BASIC programs want **F2**. Esc goes back. The Atari restarts and boots from the disk. Programs write to the .atr as to a real disk (make the file read-only on the PC to protect it).

- **Program (.xex)**: F12, choose it, **Enter**. A small loader switches BASIC off and starts the program. A program that loads into $0700-$08FF does not work this way (use an .atr version).
- **Cartridge (.car)**: F12, choose it, **Enter**: standard 8 KB and 16 KB, Atarimax 128 KB and 1 MB. Choosing an .atr or .xex takes the cartridge out.
- **Snapshots (.a8s)**: F2 saves; F12, the .a8s, Enter continues. Load it on the same model (800 / 130); for a cartridge game load the .car first.
- **Without a disk and without BASIC**: hold **Alt + F4** while you start LTari in the machine menu, until the screen appears.

### The Atari keyboard

| PC key | Atari key |
|---|---|
| Letters, digits, Return, Space, Tab, Esc, Caps Lock, Backspace | The same keys (Backspace = DELETE / BACK S) |
| - = ; , . / | The same keys |
| ' (apostrophe) | + (Shift: \) |
| \ (backslash) | * (Shift: ^) |
| [ and ] | < and > (Shift: CLEAR / INSERT line) |
| ` (back quote) | Inverse video (the Atari logo key) |
| Cursor keys | The cursor keys (Control + - = + *) |
| Home | CLEAR screen (Shift + <) |
| Delete / Insert | Delete / insert a character |
| Shift + 2 / Shift + 7 / Shift + 8 | " / ' / @ (as printed on the Atari) |

The key positions follow a US keyboard. If an .atr game hangs at start or shows a half-drawn screen, load it again with **F1** (No BASIC).

## 10. Problems

| Problem | What to do |
|---|---|
| No picture after F8 | The monitor does not accept 50 Hz: press F8 again. |
| L64: no picture for 2 seconds after starting | Normal: the picture starts when BASIC is ready. |
| The browser shows no files | The card must be FAT32 or FAT16, not exFAT. Press F12 again after changing the card. |

| Problem | What to do |
|---|---|
| A machine stays black or does not start | Install `LUPDATE.BIN` again (Installation Guide); for L64 check the C64 ROM files. |
| **Could not start** in the menu | That machine's part of the flash is damaged: install `LUPDATE.BIN` again. |
| **No answer from the ESP32** (F3) | Check that the ESP32 board sits in its sockets; switch the board off and on. |
| Keys do nothing in a machine | Close the browser or menu (F12, F3, F4 or Esc); if needed, Ctrl + Alt + Del and start the machine again. |
| Atari game hangs at start | Load the .atr again and answer **F1** (No BASIC). |